High-Performance Order Matching Engine — Project Overview

> ***46,500+ lines of C++20*** *| 85 headers | 13 source files | 79 test files | 426 CTest targets*
>
> *A C++20 low-latency matching engine drawing on institutional exchange design principles —* ***237 ns P50 core-matching latency (Clang PGO) validated on x86 Xeon bare metal*** *(≈125 ns on Apple Silicon) — with horizontal scalability.*

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1. Executive Summary

This is a C++20 low-latency order matching engine with institutional-grade architecture drawing on exchange design principles. It implements O(1) price-level lookup via `FlatPriceMap`, lock-free MPSC queues, thread-per-symbol horizontal scaling, CRC-32 journaling with deterministic replay, four wire protocols (FIX 4.2/4.4, OUCH 4.2, ITCH 5.0, SBE) over both real TCP and UDP transports, MoldUDP64 multicast with gap-recovery retransmission service, TLA+-verified safety invariants, cross-host log replication, and a complete operational stack including config management, webhook alerting, Prometheus metrics, and Docker deployment.

Codebase Statistics

Metric	Count
Total C++ LOC	~46,500
Header files (`include/`)	85
Source files (`src/`)	13
Test files	79
Individual test cases	426 CTest targets
TLA+ specifications	12 (454M+ states verified)
Documentation files	6 (plus architecture/benchmark docs)

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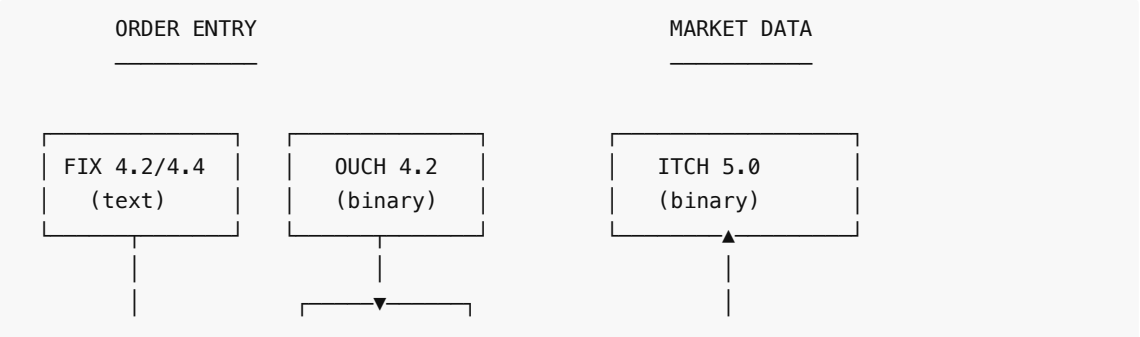
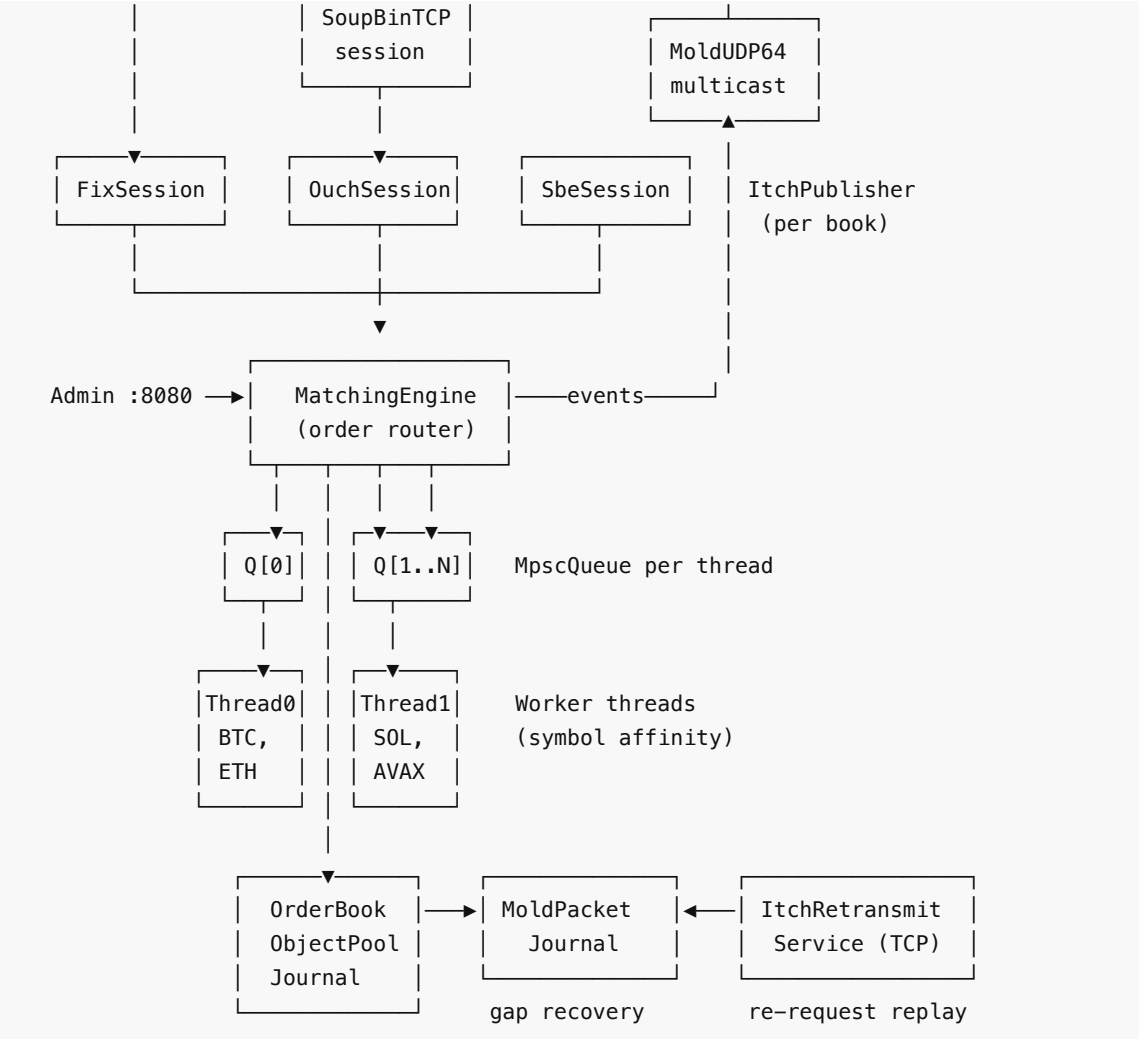
2. Core Engine Architecture

System Topology





Order Types Supported

- **Standard**: Limit, Market
- **Time-In-Force**: IOC (Immediate-or-Cancel), FOK (Fill-or-Kill), DAY, GTD (Good-Til-Date)
- **Conditional**: Stop, StopLimit, TrailingStop
- **Institutional**: Pegged, Iceberg (hidden quantity), Hidden, PostOnly, MIT, MOC, LOC
- **Match Algorithms**: Price-Time FIFO and Pro-Rata allocation with LMM/DMM floor guarantee (40% of available qty) and rounding-remainder priority

Core Data Structures

Component	Purpose	Complexity
`FlatPriceMap`	Price-level lookup by tick index	O(1) insert/lookup
`FlatHashMap`	Robin-Hood open-addressing hash map	O(1) amortized
`IntrusiveList`	Doubly-linked order queue per price level	O(1) insert/remove
`ObjectPool`	Pre-allocated slab allocator for `Order` nodes	O(1) alloc/dealloc
`RingBuffer`	Cache-line aligned lock-free ring buffer	O(1) push/pop

`MpscQueue`	Multi-producer, single-consumer lock-free queue	O(1) push/pop

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3. Concurrency & Networking

Thread-Per-Symbol Partitioning

N worker threads with independent lock-free `MpscQueue` ring buffers. Orders are deterministically routed by `hash(symbolId) % numThreads`, eliminating cross-thread contention on the hot path.

Multi-Protocol Order Entry

All four protocols dispatch into the same `MatchingEngine`.

- **FIX 4.2 / 4.4**: ASCII text + checksum, full session-layer state machine. `TransactTime` validation and version negotiation per accepted `BeginString`.
- **OUCH 4.2**: NASDAQ binary order entry, big-endian fixed-width.
- **SBE (Simple Binary Encoding)**: FIX-TG schema-driven binary (CME / ICE style). Forward and backward compatibility. Encode speed of ~1 ns/op.
- **SoupBinTCP**: TCP envelope wrapping OUCH on the wire.

Market Data Stack

- **ITCH 5.0 Publisher**: Converts engine events to ITCH 5.0 frames.
- **MoldUDP64 Multicast**: Batched publisher with MTU auto-flush + gap-detecting subscriber.
- **Gap Recovery**: Bounded ring of journaled MoldUDP64 messages backing a SoupBinTCP-over-TCP retransmission service.
- **Shared Memory IPC**: L2 market data distribution to co-located consumers via `shm_open`.

Kernel-Bypass Ingestion (DPDK / F-Stack)

Optional kernel-bypass order-entry path behind `#ifdef OB_HAVE_DPDK` (CMake `-DENABLE_DPDK=ON`): `DpdkGateway` runs F-Stack (DPDK userspace TCP over the AWS ENA PMD) on a dedicated **secondary ENI**, leaving the primary interface kernel-managed so SSH is never lost. A **single RX queue / single busy-poll lcore** preserves FIFO submit order (RSS splitting one connection across lcores would reorder and break price-time priority). F-Stack owns the NIC and runs FreeBSD's TCP stack internally; the gateway consumes its BSD-socket API (`ff_recv`), so ordered TCP payload flows straight into `OuchSession::feed()` and the engine submit path **unchanged** — DPDK replaces only how bytes arrive from the NIC, not how they are parsed. When `OB_HAVE_DPDK` is undefined (the default, and every non-Linux build) the kernel `TcpGateway` is the only ingestion path and the binary is byte-for-byte identical. **Implemented and locally verified** (macOS builds cleanly with the DPDK path `#ifdef` 'd out; gateway tests green); **AWS validation pending** via `scripts/dpdk_aws.sh`.

Wire-to-Wire Latency Measurement

A two-host OUCH-over-TCP harness — `tools/WireLatencySender` (instance A) and `tools/WireLatencyReceiver` (instance B) — measures NIC-to-NIC order-entry latency. On Linux it captures NIC **hardware** timestamps via `SO_TIMESTAMPING` (TX from the socket error queue, RX from the `recvmsg` cmsg), with a `CLOCK_MONOTONIC` **software fallback** on non-Linux / non-HW hosts (each CSV row records which source it used). One-way latency is estimated as **round-trip / 2**, which sidesteps clock synchronisation entirely — both timestamps are on the sender's own clock, so no PTP is needed for a first-pass number. The receiver also supports the F-Stack path (`--conf`) so the DPDK and kernel sessions emit directly comparable CSV, and a `--software-timestamps` flag forces both onto an identical

CLOCK_MONOTONIC basis. **Implemented and locally verified** (both tools build on macOS; loopback smoke test passes); **AWS validation pending** via `scripts/wire_latency_aws.sh`.

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4. Regulatory & Risk Management

Feature	Implementation
Self-Match Prevention (SMP)	Cancel-Taker strategy — prevents wash trading
Circuit Breakers	Configurable % price-band breach enters a short LULD-style **volatility auction** (orders accumulate without continuous matching until the reopening uncross) rather than a hard halt; emits a `breaker_trip` audit event
OTR Monitoring	Real-time Order-to-Trade ratio tracking per participant
Kill Switch	Instant cancellation of all orders per participant across all symbols
Pre-Trade Risk Limits	Max order size, notional value, and position limits per participant
Rate Limiting	Token-bucket throttling at ingress (configurable rate + burst); `RateLimiter::reconfigure()` live-updates default rate/burst and clears all participant buckets — called on SIGHUP
Queue Backpressure	Rejects orders when queue exceeds configurable threshold (default 80%)
LMM/DMM Privileges	`ParticipantRole` enum; ProRata 40% floor guarantee to LMM/DMM orders; remainder priority

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5. Microstructure Research Infrastructure

A self-contained quantitative research layer that runs against the live matching engine, enabling empirical microstructure analysis and strategy development.

Component	Purpose
`SimulationDriver`	Multi-agent market simulator (NoiseTrader, MarketMaker, InformedTrader)
`MicrostructureMetrics` / `ResearchHarness`	Snapshot collection and research session management
`VPINCalculator`	Volume-Synchronized Probability of Informed Trading
`SpreadDecomposition`	Huang-Stoll and Glosten-Harris decomposition (adverse selection, inventory, order processing components)
`OrderFlowAnalytics`	Roll spread, queue imbalance signal, logistic fill-probability model
`ImpactModel`	Almgren-Chriss + square-root market impact laws

`ExecutionAnalytics` / `OptimalExecution`	Almgren-Chriss execution schedule, arrival-price and VWAP slippage
`CalibrationPipeline`	Nelder-Mead minimization of spread decomposition residuals
`BacktestEngine`	Historical tape replay against research models
`SignalGenerator`	Multi-factor composite signal (momentum, spread, VPIN, imbalance)
`PaperTrader`	Signal-driven IOC order submission with position and P&L tracking
`ResearchDashboard` / `ResearchSerializer`	Result visualization and persistence

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6. Reliability & High Availability

CRC-32 Journaling

- Write-ahead log with atomic CRC-32 integrity on every entry.
- Crash recovery: replay stops at first corrupted/truncated record.
- Atomic checkpoint: snapshots active book state, rewrites journal.
- Deterministic replay via virtual clock (`setExpiryClock(ClockFn)`).
- Auto-rotation: `OB_JOURNAL_MAX_SIZE_MB` env var triggers `engine.checkpoint()` from the main loop when `Journal::needsCheckpoint()` fires.

Cross-Host Log Replication (wired end-to-end)

- `ReplicationCoordinator` — TCP log-shipping from primary to backup, instantiated in `src/main.cpp` driven by `OB_NODE_ROLE` / `OB_PRIMARY_HOST` / `OB_JOURNAL_PATH` env vars.
- `Journal::onCommit` hook ships each fsync-durable batch to the backup; backup's `applyReplicatedEntry` writes to its own journal so the replica is durable across its own restarts.
- `HeartbeatMonitor` — configurable failure detection timeout.
- `LeaderLease` + `LeaseGrant` propagation — primary broadcasts lease state every heartbeat tick; backup's `tryAcquire` is fenced on local lease expiry, not just heartbeat miss. Prevents split brain under partial network failure (verified by 19-scenario chaos suite: partition, asymmetric partition, 30% packet loss, +30s clock skew — 0 split brain in all).
- Transport auto-reconnect — `receiveLoop` re-runs `connectTo()` against saved host/port after socket loss; ~3 ms recovery in chaos tests. Reconnect uses exponential backoff: 500ms → 30s cap, resets to base on successful connect.
- Snapshot catchup on join — primary streams every resting order via `MatchingEngine::streamSnapshot` when a backup connects; idempotent on receiver.
- Automatic backup promotion on combined heartbeat-miss + lease-expiry; `setReplayModeAllBooks(false)` flips backup into a live primary.

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7. Observability & Operations

Observability Features

- **Structured Logging**: Pluggable `StructuredSink` API (Null, JSON Stderr, Capturing). 7 typed `obSink().log()` events wired in `OrderBook.cpp`

(accepted/cancelled/rejected/fill/breaker/state); backends hot-swappable via typed helpers in `StructuredLog.h`.

- **Prometheus Metrics**: `MetricsRegistry` with `/prometheus` text-exposition endpoint. Tracks order flow, queue depth, latencies, and journal health.
- **Webhook Alerting**: `AlertDispatcher` with targets for Slack, PagerDuty, or HTTP webhooks.

Admin HTTP Server (Port 8080)

Endpoint	Purpose
`GET /health`	K8s liveness probe
`GET /readyz`	K8s readiness probe — HTTP 503 until warmup, then 200; auth-exempt; separate from liveness `/health`
`GET /metrics`	Internal counters (JSON)
`GET /prometheus`	Prometheus text exposition
`GET /book?symbolId=0`	L2 order book snapshot
`GET /otr?participantId=1`	Order-to-trade ratio

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8. Performance Benchmarks

`HonestBenchmark` feeds one deterministic order flow (50K orders, seed=42) through three cumulative paths. The **x86 figures are the authoritative, reproducible numbers**, validated on AWS bare metal; Apple Silicon dev-machine numbers follow as a reference. P50 is the stable per-operation figure; throughput is wall-clock and load-sensitive. (Per-order/per-fill structured logging and event dispatch are sink-/listener-gated, so the hot path stays allocation- and vtable-free when nothing is attached.)

Validated x86 — AWS c6in.metal (authoritative, standard Release build)

Dual-socket Intel Xeon Platinum 8375C @ 2.90 GHz, hyperthreading disabled (`nosmt`), Ubuntu 26.04, Clang C++20 `-O3 -march=native` , `numactl --cpunodebind=0 --membind=0` . 5 stable runs, post-optimization commit `d2e688c` .

Path	What's Included	P50	P90	P99	Throughput
Core matching	OrderBook + STP + WashTrade + LULD	**261 ns**	620 ns	1,010 ns	2.80M ops/s
Engine wrapper	+ sequence alloc, rate limiter	269 ns	620 ns	1,001 ns	2.74M ops/s
Full-stack journal	+ GroupCommit (batch=64, async io_uring ack on EBS)	615 ns	1,048 ns	3,568 ns	1.28M ops/s

PGO: Clang IR-based profile-guided optimization (profiled on the seed=42 HonestBenchmark workload) takes Path A core matching to **P50 237 ns / P99 910 ns / 3.10M ops/s** — the headline figure. The table above is the standard (non-PGO) Release build.

`perf` (Path A, 50K orders seed=42): IPC 1.34 · 17.8 branch-misses/order · 152 L1-dcache-misses/order · 12,638 instructions/order.

Apple Silicon (M-series dev machine, reference)

Path	P50	Note
Core matching	~125 ns	~42 ns clock granularity quantizes per-path P50s, so Path A/B can read equal or invert run-to-run
Engine wrapper	~125 ns	
Full-stack journal	~1,400 ns	macOS/APFS `fdatasync` artifact — **not structural** (the same path is 615 ns on Linux x86, async io_uring ack)

The ~2.2× ARM-vs-x86 gap on core matching is microarchitectural (wider out-of-order window + stronger branch prediction on pointer-chasing code), **confirmed not** caused by build flags, field ordering, branch hints, or branchless selection.

Where the 261 ns goes (and what doesn't move it)

The four earlier micro-fixes (listener-dispatch guard, `shared_mutex`→plain `mutex`, OCO scratch-buffer reuse, rehash guard) produced **no measurable x86 latency change** — the P50 is **structurally bound**, not instruction-bound (17.8 branch-misses/order, 152 L1-dcache-misses/order). This cycle's branchless price-cross *did* move it, shaving 10 ns P50 / 62 ns P99 to reach 261 ns (see Optimization History in BENCHMARKS.md); next-order prefetch and the price-level arena allocator were both implemented and reverted as net-negative on this 100%-fill flow.

Cost	ns	Driver	Lever (estimated)
Pointer chasing (intrusive list)	80–100	152 L1-dcache misses/order	arena allocator (reverted — net-negative)
Branch mispredicts	60–80	17.8/order, data-dependent	branchless price-cross (shipped, –10 ns P50)
Irreducible work	50–60	price/qty math, STP, compliance	—
Spectre mitigation (eIBRS)	30–40	kernel-enforced on this instance	not disableable here

Confirmed **0 ns delta** on this workload: `-O2` vs `-O3`, `Order` field reordering, `[[likely]]` / `[[unlikely]]` hints, branchless `isBuy` book selection. Shipped this cycle: branchless price-cross (–10 ns P50 / –62 ns P99) and io_uring async journal ack (Path C P99 5,312→3,568 ns); next-order prefetch and the price-level arena allocator were both reverted as net-negative. Clang IR-based PGO then took Path A to 237 ns P50 (the headline figure). The journal P99 (~3.6 µs) is the `fdatasync`/EBS flush; NVMe/RAM-backed storage would be materially lower.

Binary Codec Microbenchmark

Codec	Message	ns/op	M ops/s

OUCH 4.2	EnterOrder encode (49B)	112.0	8.9
ITCH 5.0	AddOrder encode (36B)	34.1	29.4
SBE	**NewOrderV1 encode (32B)**	**1.0**	**1015**
SBE	**NewOrderV1 decode (32B)**	**0.5**	**2128**

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9. Verification & Testing

Test Suite — 79 Executables, 426 CTest Targets

The testing infrastructure includes Unit, Functional, Integration, Chaos, Property, Shadow, and Benchmark testing categories across 79 test executables and 426 CTest targets. Key mechanisms:

- **Shadow Mode**: Dual-book divergence detection, validating FIFO compliance.
- **Fault Injection**: 10+ injection points (short-writes, pool exhaustion, EAGAIN injection) with zero-cost overhead in production.
- **Coverage-Guided Fuzzing**: libFuzzer harness for protocol parsing and order flow.
- **Sanitizers**: ASan, UBSan, and TSan checks integrated into CI/CD.

Formal Verification (TLA+)

12 TLA+ specifications, model-checked with TLC:

- **`MatchingEngine.tla`**: 454M states verified. Zero violations for `NoNegativeQuantity`, `FIFO_Preservation`, and `GTD_Expiry_Correctness`.
- **`Replication.tla`**: Realistic lease-propagation model (no god-mode `~primaryAlive` guard on `BackupPromote`; promotion requires both heartbeat-miss AND local-lease-expiry). 1373 → 4192 states verified at `MaxEntries=6 → 10`, zero violations. A bug-injected variant (lease check stripped) reproduces split brain in 188 states, confirming the verification is genuine.
- **`MpscQueue.tla`**: Lock-free ring buffer linearizability.
- **`SnapshotLocked.tla`**: Mutex torn-snapshot prevention.

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10. File Structure

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include/              85 header files — core logic and networking
src/                  13 source files — thin compilation units
tests/                79 test files, 426 CTest targets
benchmarks/           9 benchmark binaries
fuzz/                 9 files — coverage-guided fuzzing harnesses
spec/                 TLA+ formal specifications (12 specs)
tools/                CLI tools and data utilities
config/               Example configuration files
docs/                 Runbooks, CapacityPlanning, ProductionReadiness
```

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11. Remaining Work

The system is architecturally complete. The remaining items are AWS validation steps for capabilities already implemented and locally verified (nothing is hardware-blocked — the kernel-bypass path runs on

commodity AWS ENA hardware):

- **x86 Bare Metal Benchmarks**: ✅ done — validated on AWS c6in.metal (dual Xeon 8375C, `nosmt`, NUMA-pinned); see §8. Core matching 261 ns P50, confirming the structural-bottleneck analysis (pointer-chasing L1 misses + data-dependent branch mispredicts + Spectre eIBRS). The prior four micro-optimizations moved x86 latency 0 ns; this cycle's branchless price-cross shaved 10 ns while next-order prefetch and the arena allocator were both reverted as net-negative — confirming the P50 is structurally bound.
- **Journal Async I/O (io_uring)**: ✅ done — validated on x86 Linux (AWS c6in.metal). The async ack (onCommit fires on the completion-reaper thread, not on submit) cut Path C P99 **32.8%** (5,312 → 3,568 ns) at the cost of +167 ns P50. io_uring is a generic Linux async-I/O interface and needs no special NIC — only `liburing` on a modern kernel; the seam stays behind `#ifdef __linux__` with an `fdatasync` / `F_FULLFSYNC` fallback elsewhere.
- **Kernel Bypass Networking (DPDK / F-Stack)**: ⏳ implemented locally, pending AWS validation. `DpdkGateway` integrates F-Stack (DPDK userspace TCP over the AWS ENA PMD) on a secondary ENI behind `#ifdef OB_HAVE_DPDK`, feeding `OuchSession::feed()` unchanged (see §3). The integration is complete and builds locally with the path `#ifdef`'d out; `scripts/dpdk_aws.sh` provisions the receiver (hugepages, DPDK + F-Stack install, vfio-pci bind, F-Stack config). It runs on **commodity AWS ENA** hardware — no Solarflare/Onload — needing only a secondary ENI, hugepages, and the DPDK/F-Stack toolchain; the two-instance c6in.metal run is the next step.
- **`Replication.tla` Verification**: ✅ done — see Verification section above. The original "not yet verified" footnote is obsolete.
- **Wire-to-Wire Latency Validation**: ⏳ harness implemented locally, pending AWS validation. The `WireLatencySender` / `WireLatencyReceiver` harness (`SO_TIMESTAMPING` hardware timestamps with `CLOCK_MONOTONIC` software fallback, round-trip/2 one-way estimate — see §3) and `scripts/wire_latency_aws.sh` are ready; the remaining step is a two-instance AWS c6in.metal run (placement group, secondary ENI).

Developed for professional quantitative trading systems. C++20 · ~46,500 LOC · 79 test executables · 426 CTest targets · 19 multi-container chaos scenarios · 454M TLA+ states verified on MatchingEngine.tla · 12 TLA+ specifications